



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 3

1. **Title:** Implement control structures and form validation; create a grading system based on user-entered marks.

2. **Software/Tools Required:** VS Code, web browser, HTML, JavaScript.

3. **Theory: Control structures** (`if`, `else if`, and `else`) are used to make decisions based on the marks entered by the user.

Form validation checks whether the input is empty or outside the valid range (0–100) before processing.

The `getElementById()` method is used to access the value entered in the input box.

The `Number()` function converts the entered value from text to a numeric value. A **function** (`calculateGrade()`) is called when the **Submit** button is clicked to validate the input and calculate the grade.

The calculated grade is displayed on the webpage using the `innerHTML` property.

4. Output :

The screenshot shows a web browser window with the address bar displaying the file path: `/Users/shreyamaske/jsexp3/Shreya1196exp3.html`. The page has a purple background and features the title "Shreya's 1196 Grading System" in a large, bold, black font. Below the title, the text "Enter Marks:" is displayed. Underneath, there is a text input field with the placeholder text "Enter Marks here". Below the input field is a button labeled "Submit".



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The screenshot shows a web browser window titled "Grading System" with a file path in the address bar: `/Users/shreyamaske/jsexp3/Shreya1196exp3.html`. The page has a dark purple background and displays the title "Shreya's 1196 Grading System". Below the title, it says "Enter Marks:" followed by a text input field containing the number "106". A "Submit" button is located below the input field. A white error message box is overlaid on the right side of the page, stating "This page says Enter Valid Marks between 0 and 100" with an "OK" button.

This screenshot is similar to the previous one, showing the same web application. However, the text input field now contains the placeholder text "Enter Marks here". The error message box is still present, but the message has changed to "This page says Enter Marks" with an "OK" button.

The final screenshot shows the web application with a light purple background. The title "Shreya's 1196 Grading System" is at the top. Below it, it says "Enter Marks:" followed by a text input field containing the number "74". A "Submit" button is below the input field. Below the "Submit" button, the results are displayed: "Marks: 74" and "Grade: B+".

5. Result/Conclusion: The grading system was successfully implemented using JavaScript. The program validated the user input, assigned grades using if-else control structures, and displayed the correct grade based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



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Case Study No.: 3

1. **Case Study Title:** Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

Write a JavaScript program that accepts student marks and display the Scholarship category.

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Scholarship Eligibility</title>
</head>
<body style="background-color:rgba(176, 230, 187, 0.867);">
  <h1>SIT Scholarship Eligibility</h1> <hr><hr>
  <label>Enter Marks:</label><br><br>
  <input type="text" id="marks" placeholder="Enter Marks
here"><br><br>
  <button onclick="calculateEligibility()">Submit</button>
  <h2 id="result"></h2>
  <script>
    function calculateEligibility() {
      let marks=document.getElementById("marks").value;
      if(marks == "")
      {
        alert("Enter Marks");
        return;
      }
    }
  </script>

```



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```
marks = Number(marks);  
if(marks < 0 || marks > 100)  
{  
    alert("Enter Valid Marks for Eligibility");  
    return;  
}  
let grade;  
if(marks >= 95)  
    grade = "Platinum Scholarship";  
else if(marks >= 85)  
    grade = "Gold Scholarship";  
else if(marks >= 70)  
    grade = "Silver Scholarship";  
else if(marks >= 50)  
    grade = "Bronze Scholarship";  
else if(marks <= 50)  
    grade = "Not Eligible for Scholarship";  
document.getElementById("result").innerHTML =  
    "Marks: " + marks + "<br>Grade: " + grade;  
}  
</script>  
</body>  
</html>
```



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3. Output:

Screenshot of the SIT Scholarship Eligibility form. The browser window shows the URL `/Users/shreyamaske/jsexp3/shreya1196case3.html`. The form has a green header with the title **SIT Scholarship Eligibility**. Below the title, it says **Enter Marks:** followed by a text input field containing `Enter Marks here` and a `Submit` button.

Screenshot of the SIT Scholarship Eligibility form with a validation message. The browser window shows the URL `/Users/shreyamaske/jsexp3/shreya1196case3.html`. The form has a dark green header with the title **SIT Scholarship Eligibility**. Below the title, it says **Enter Marks:** followed by a text input field containing `Enter Marks here` and a `Submit` button. A validation message box is displayed over the form, stating **This page says** `Enter Marks` with an `OK` button.

Screenshot of the SIT Scholarship Eligibility form showing the result. The browser window shows the URL `/Users/shreyamaske/jsexp3/shreya1196case3.html`. The form has a green header with the title **SIT Scholarship Eligibility**. Below the title, it says **Enter Marks:** followed by a text input field containing `32` and a `Submit` button. Below the form, the text **Marks: 32** and **Grade: Not Eligible for Scholarship** is displayed.



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Scholarship Eligibility

File /Users/shreyamaske/jsexp3/shreya1196case3.html



SIT Scholarship Eligibility

Enter Marks:

Submit

Marks: 86
Grade: Gold Scholarship

4. Result/Conclusion: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.